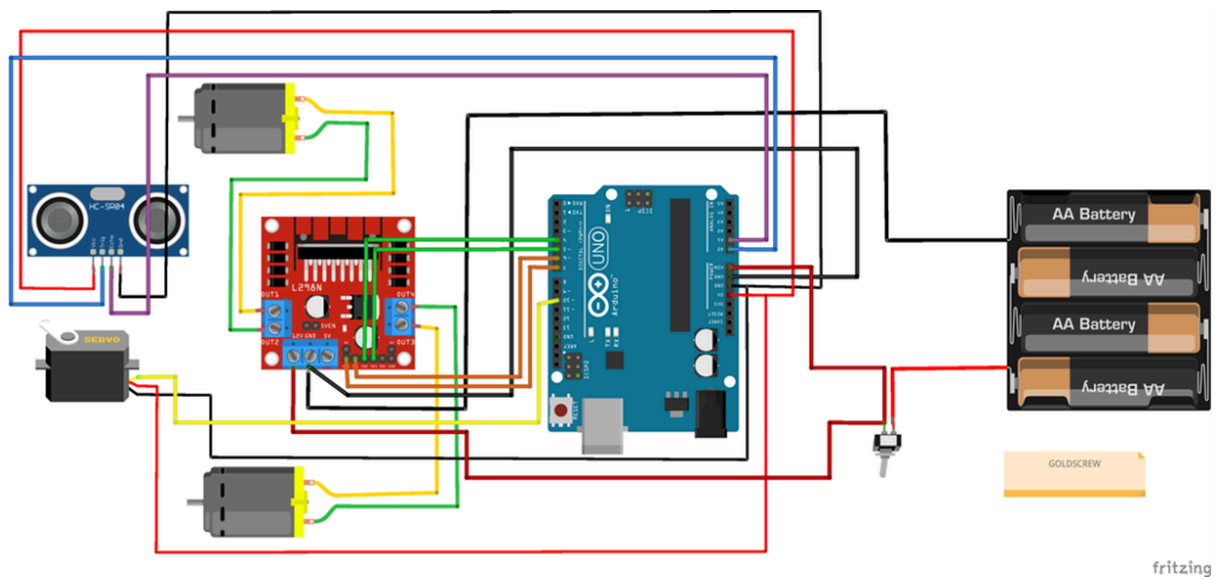


4 WHEEL CAR + ULTRASONIC SENSOR

- ASSEMBLY



- CODE

```
// First Include the NewPing and Servo Libraries

#include <NewPing.h> // library uses to read more accurate values from
ultrasonic sensor
#include <Servo.h>

#define TRIG_PIN A0
#define ECHO_PIN A1
#define MAX_DISTANCE 200
NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE); // sensor function
Servo myservo;
int distance = 100;
// Motor A
const int motorPin1 = 8; //left side 1st pin
const int motorPin2 = 9; //left side 2st pin

//Motor B
const int motorPin3 = 10; //right side 1st pin
const int motorPin4 = 11; //right side 2st pin

void setup() {

    myservo.attach(5); // pin of servo motor
    myservo.write(115);
    delay(2000);
    distance = readPing();
    delay(100);
    distance = readPing();
    delay(100);
    distance = readPing();
    delay(100);
    distance = readPing();
    delay(100);

}

void loop() {
    int distanceR = 0; // right
    int distanceL = 0; // left
    delay(40);
```

```
if(distance<=20)
{
    moveStop();
    delay(100);
    moveBackward();
    delay(300);
    moveStop();
    delay(200);
    distanceR = lookRight();
    delay(200);
    distanceL = lookLeft();
    delay(200);

    if(distanceR>=distanceL)
    {
        turnRight();
        moveStop();
    }
    else
    {
        turnLeft();
        moveStop();
    }
}
else
{
    moveForward();
}

distance = readPing();
}

int lookRight()
{
    myservo.write(50);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(115);
    return distance;
}
```

```

int lookLeft()
{
    myservo.write(170);
    delay(500);
    int distance = readPing();
    delay(100);
    myservo.write(115);
    return distance;
    delay(100);
}

int readPing() { // defines a function returns an integer
    delay(70); // delay of 70 milliseconds
    int cm = sonar.ping_cm(); //The ping_cm() function is used to
measure the distance in centimeters for ultrasonic.
    if(cm==0) // if the measurement is 0 (which could
indicate a problem), it adjusts the value to 250 before returning it.
    {
        cm = 250;
    }
    return cm;
}

void moveStop() {
    analogWrite(motorPin1, 1);
    analogWrite(motorPin2, 1);
    analogWrite(motorPin3, 1);
    analogWrite(motorPin4, 1);
}

void moveForward() {

    analogWrite(motorPin1, 180);
    analogWrite(motorPin2, 0);
    analogWrite(motorPin3, 0);
    analogWrite(motorPin4, 180);
    // digitalWrite(motorPin4, HIGH);

}

void moveBackward() {

```

```
    analogWrite(motorPin1, 0);
    analogWrite(motorPin2, 180);
    analogWrite(motorPin3, 180);
    analogWrite(motorPin4, 0);
}

void turnRight() {
    analogWrite(motorPin1, 180);
    analogWrite(motorPin2, 0);
    //stop right side to move right
    analogWrite(motorPin3, 0);
    analogWrite(motorPin4, 0);
    delay(300);
    moveForward();
}

void turnLeft() {
    analogWrite(motorPin1, 0);
    analogWrite(motorPin2, 0);

    analogWrite(motorPin3, 180);
    analogWrite(motorPin4, 0);
    delay(300);
    moveForward();
}
```